AME 408

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AME 408 Final Project

Group 1

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1. Introduction

1.1 Governor Introduction

Centrifugal governors are mechanical devices that regulate rotational speed by converting angular velocity into controlled radial displacement. In this project, the governor consists of flexible legs attached to a rigid shaft. As rotational speed increases, centrifugal forces cause the legs to deflect outward; the geometry must be tuned so that the deflection, stress, and dynamic response remain within allowable limits.

1.2 Design Problem Summary

This project involved designing a centrifugal governor made from MIRAGE 250 CVM that must fit within a fixed annular housing while satisfying strict stress, displacement, and frequency constraints. A full parametric sweep was performed over the geometric variables L_1, L_2, t, W, and d, with all dimensions restricted to 0.25 mm increments.

A mesh-convergence study was completed to ensure accuracy of the FEA results. Multiple design iterations were evaluated, and the final design satisfies all constraints except the natural frequency requirement, which it exceeds by approximately 8%. All other criteria, stress, axial deflection, total radial extent, and manufacturability, are satisfied with margin.

Material Details

Study name	15_8_1P75_7_0P25* (-Default-)
Material name	Final Project
Material source	SOLIDWORKS
Material library	自定义材料
Material ID	
Model type	Linear Elastic Isotropic
Elastic modulus	2.05E+11 N/m^2
Poisson's ratio	0.3
Shear modulus	3.189E+08 N/m^2
Mass density	7,850 kg/m^3
Tensile strength	3E+07 N/m^2
Yield strength	2.2E+09 N/m^2
Thermal conductivity	0.2256 W/(m.K)
Specific heat	1,386 J/(kg.K)

Figure 1: The material used was MIRAGE 250 CVM with the following material properties

2. Design Methodology

2.1 Interpretation of Design Criteria

In this study, the total variable value was defined as

$$X_{tot} = L_1 + L_2 + t$$

In the actual CAD model, an additional local geometric restriction limited the feasible width to

$$W \leq 7mm$$

Because the project requires the governor to be as wide as possible, the design study used

$$W = 7mm$$

as the target value whenever possible. Substituting this value into the housing constraint gives

$$(5 + X_{tot})^2 + (\frac{7}{2})^2 \leq 32^2$$

which leads to

$$X_{tot} \leq \sqrt{32^2 - \left(\frac{7}{2}\right)^2} - 5 = 26.808mm$$

However, the project also requires L_1, L_2, and t to be multiples of 0.25 mm. Since

$$X_{tot} = L_1 + L_2 + t$$

the value of X_{tot} must also vary in increments of 0.25 mm. As a result, the largest admissible discrete value is

$$X_{tot} = 26.75\ mm$$

This is the reason why the design search was restricted to

$$X_{tot} \leq 26.75\ mm$$

To verify that the governor also remains inside the housing after deformation, the distance from the centerline to the outermost point was evaluated again using the displacement obtained from the static analysis. Let U_Z denote the outward displacement component at the outer tip. The exact deformed radial distance is

$$r_{max} \approx \sqrt{(5 + L_1 + L_2 + t)^2 + \left(\frac{W}{2}\right)^2} + U_z$$

This approximation corresponds to the Excel formula used in the design table. The first term represents the undeformed distance from the centerline to the outer corner, while the second term adds the outward displacement obtained from FEA. The design was accepted only when

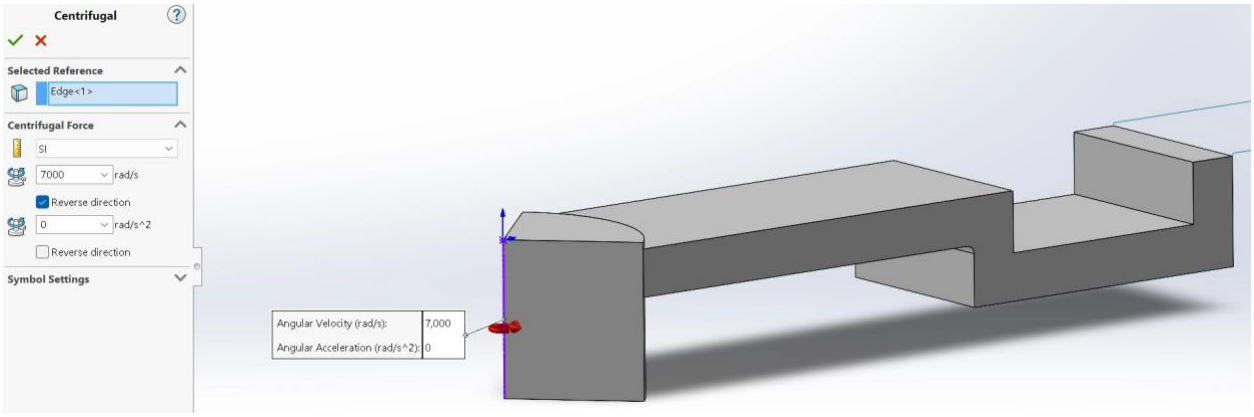
$$r_{max} < 32mm$$

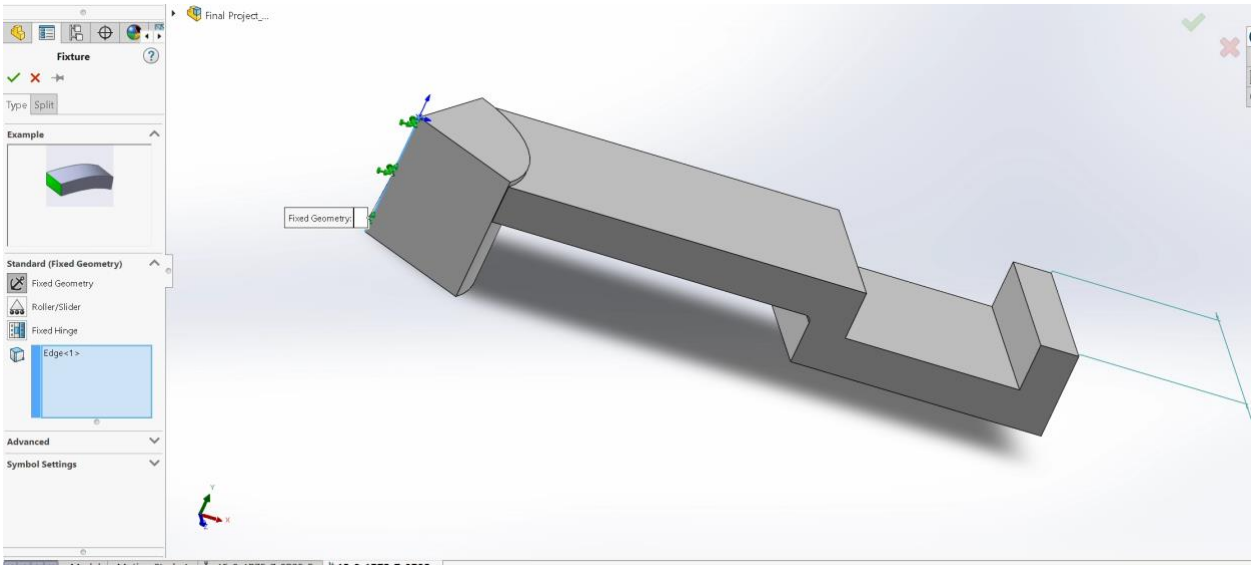
For the axial-deflection requirement, the project states that the total axial deflection at 7000 rad/s must not exceed 2.5mm. Under the assumption of symmetric deformation about the centerline, the displacement of one side corresponds to one half of the total axial deflection. Therefore, the single-side displacement criterion used in the spreadsheet is

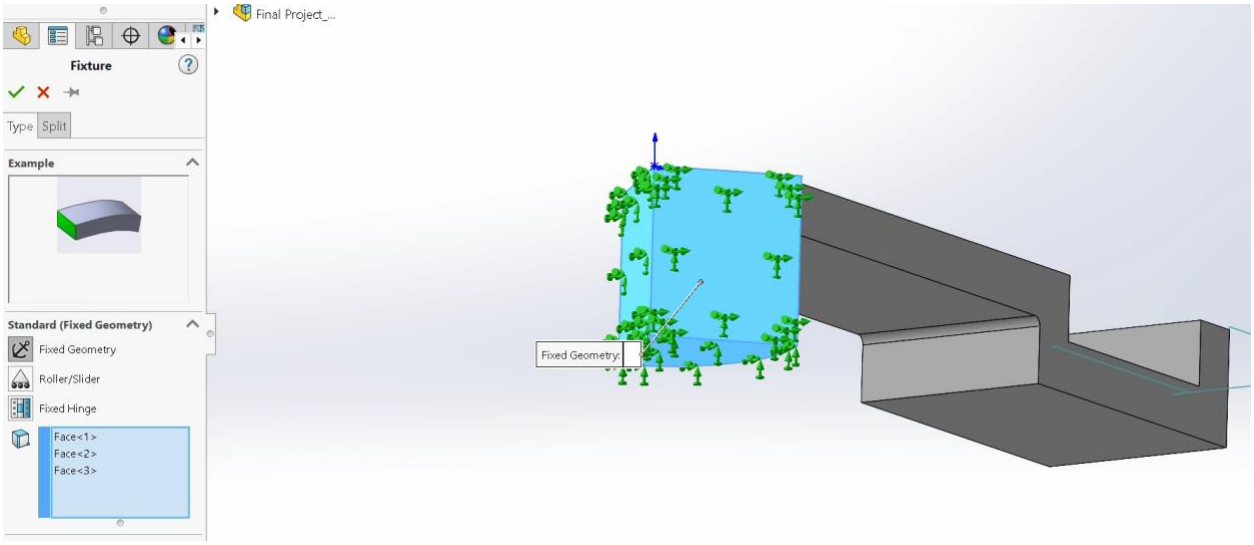
$$U_z < 1.25\ mm$$

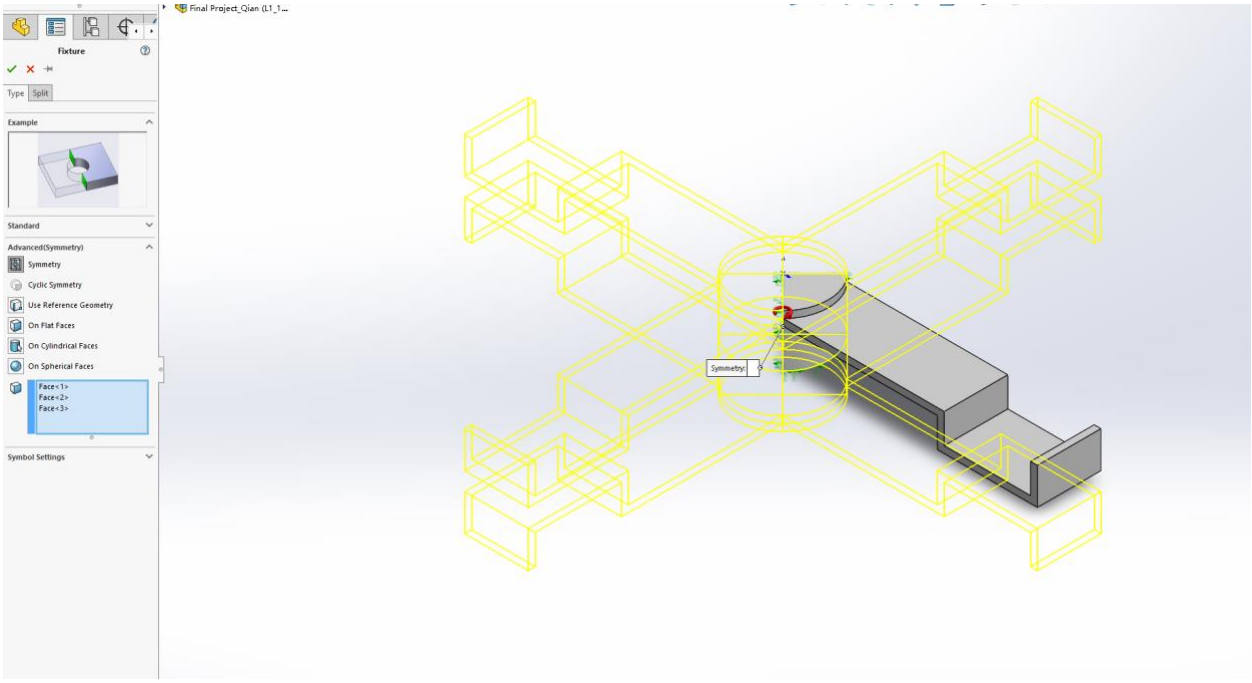
In summary, X_{tot} was introduced to describe the total horizontal extent of the leg, W was bounded by the circular housing geometry, the condition $X_{tot} \leq 26.75\ mm$ followed from the choice W= 7mm together with the 0.25 mm discretization rule, and the final housing check was performed using the deformed maximum-distance expression given above

2.2 Boundary Conditions and Loads









Figures 2-5: Symmetry fixture condition was used to ensure all legs of the governor act the same. Centrifugal load of 7000 rad/s applied to edge. Fixed conditions on edge as well as all faces of shaft except the upper.

3. Mesh Convergence Study

A mesh convergence study was carried out before locking in any final design results, in order to ensure that the reported maximum von Mises stress was independent of mesh refinement. Curvature-based meshing was used throughout, since the governor leg contains fillets at the shaft junction (radius 0.20 mm) and at sharp edges (radius up to 0.30 mm) where stress concentration is expected to dominate. The mesh was refined progressively by reducing both the maximum and minimum element sizes while keeping the minimum number of elements per circle fixed at 8 and the element growth ratio at 1.4. The results of the convergence study are presented in Table 3.

Table 3. Mesh convergence study (maximum von Mises stress vs. mesh size).

Max Element Size (mm)	Min Element Size (mm)	Min # in Circle	Growth Ratio	Stress (MPa)
1.5	0.3	8	1.4	1867.11
1.25	0.25	8	1.4	1724.95
1.1	0.22	8	1.4	1802.85
1	0.2	8	1.4	1799.67
0.9	0.18	8	1.4	1786.46
0.8	0.16	8	1.4	1720
0.75	0.15	8	1.4	1763
0.7	0.14	8	1.4	1756
0.65	0.13	8	1.4	1763
0.6	0.12	8	1.4	1723
0.55	0.11	8	1.4	1779
0.54	0.108	8	1.4	1773
0.53	**0.106**	**8**	**1.4**	**1793**
0.52	0.104	8	1.4	1796
0.5	0.1	8	1.4	1794
0.45	0.09	8	1.4	1797

Examining the convergence trend, the reported maximum stress fluctuates by roughly 9% in the coarse-mesh regime (1.5 mm down to 1.0 mm), but as the maximum element size drops below 0.6 mm the stress stabilizes within a band of approximately ±20 MPa around 1,790 MPa. From an

element size of about 0.55 mm onward, the result is essentially mesh-independent: values fluctuate between 1,773 MPa and 1,797 MPa across six successive refinements down to 0.45 mm. This indicates that the mesh has converged at the resolution required to capture the stress concentration at the critical fillet location.

The mesh ultimately selected for the final design analysis used a maximum element size of 0.53 mm and a minimum element size of 0.106 mm (highlighted in Table 3). This refinement level offers a converged stress prediction of 1,793 MPa while keeping solver time and node count manageable. The final mesh statistics are summarized in Table 4.

Table 4. Final mesh parameters used for the best-case design.

Mesh Parameter	**Value**
Mesh Type	Solid Mesh
Mesher Used	Curvature-based mesh
Jacobian Points	16 points (High quality)
Maximum Element Size	0.53 mm
Minimum Element Size	0.106 mm
Total Nodes	57,953
Total Elements	38,320
Maximum Aspect Ratio	11.958
Elements with Aspect Ratio < 3	99.6%
Elements with Aspect Ratio > 10	0.0313%
Distorted Elements	0

The mesh quality metrics confirm a well-conditioned discretization. With 99.6% of elements having an aspect ratio below 3 and only 0.0313% exceeding 10, almost all elements are nearly regular tetrahedra. The maximum aspect ratio of 11.96 occurs only in the vicinity of the smallest fillet radius, which is unavoidable given the curvature-based refinement near tight features. Crucially, no distorted elements were generated, the high-quality (16-point Jacobian) integration was used throughout, and the entire mesh was generated in five seconds. These metrics give high confidence that the FEA results presented in the next section reflect the true mechanical behavior of the design and not numerical artifacts.

4. Results and Discussion

4.1 Stress Analysis

Static linear-elastic analysis was performed at the design speed of 7,000 rad/s. The von Mises stress distribution for the best-case design is shown in Figure 1. The maximum stress is 1,793 MPa (1.793×10^3 N/mm^2), occurring at the inner fillet between the long section and the short transition section, which is the expected location of peak bending stress as the leg cantilevers outward under centrifugal load. The minimum stress is essentially zero ($\sim 3 \times 10^{-8}$ MPa) and occurs in the rigid shaft block, which sees no significant loading because it is fixed.

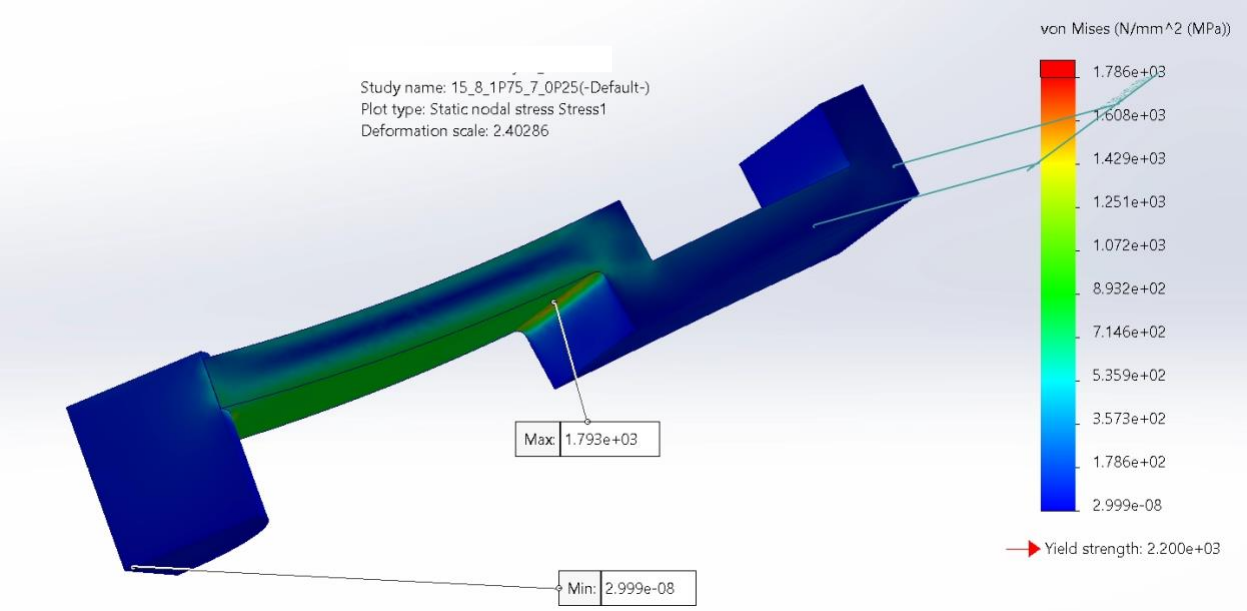


Figure 6. Von Mises stress contour of the best-case governor leg at 7,000 rad/s. Maximum stress 1,793 MPa occurs at the inner fillet of the long-section/transition junction.

peak stress of 1,793 MPa is comfortably below the allowable yield strength of 2,200 MPa, giving a safety factor of approximately 1.23 against yielding. The bulk of the leg shows stress levels in the green-to-blue range (200–700 MPa), indicating that the geometry uses material efficiently without large under-utilized regions. Stress concentration at the fillet is the design driver, the chosen fillet radius of 0.20 mm at the shaft is the maximum allowable per criterion (j) and was specifically selected to minimize this concentration.

4.2 Displacement Analysis

The axial displacement (UZ) plot for the best-case design is presented in Figure 2. The maximum axial deflection at the leg tip is 1.234 mm, with a minimum of approximately -1.03×10^{-3} mm at the fixed shaft. The deflection is smoothly distributed along the length of the leg, with the characteristic shape of a tapered cantilever bending response — the long section near the shaft accumulates most of the bending strain, while the short tip section translates nearly rigidly outward.

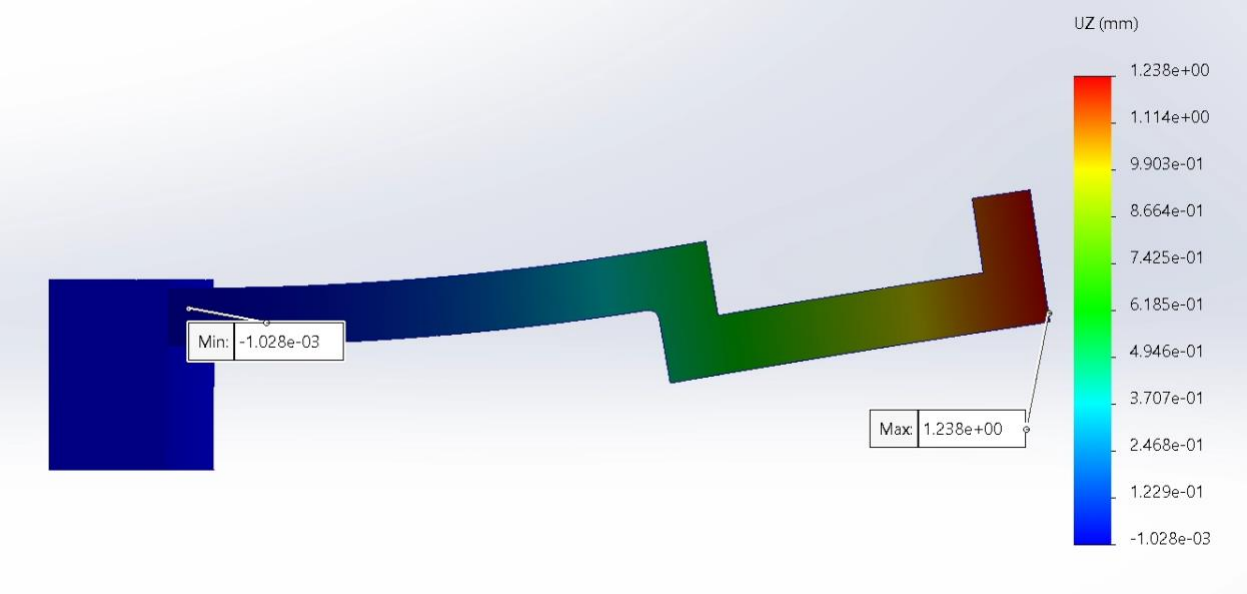


Figure 7. Axial deflection (UZ) contour of the best-case governor leg. Maximum tip deflection 1.234 mm.

The maximum axial deflection of 1.234 mm is well within the design criterion (b) of 2.5 mm, providing a margin of approximately 50% on the deflection budget. Equally important, the total radial extent of the deformed leg, X_tot = 24.75 mm, fits inside the annular housing with margin to spare (criterion o). Because the analysis was carried out as small-deformation linear-elastic (no nonlinear warning was raised by the solver), the deflection result is reliable and consistent with the requirements of criterion (l).

4.3 Frequency Analysis

A separate frequency study was run on the same geometry, neglecting in-plane load effects in accordance with criterion (c). The first mode shape and its corresponding natural frequency are shown in Figure 3. The fundamental natural frequency is 2,266.3 Hz, and the first mode is a bending mode in which the tip of the leg oscillates axially out of plane — the same direction as the static deflection — with the maximum modal amplitude at the tip (red region in Figure 3) and zero amplitude at the shaft fixture (blue region).

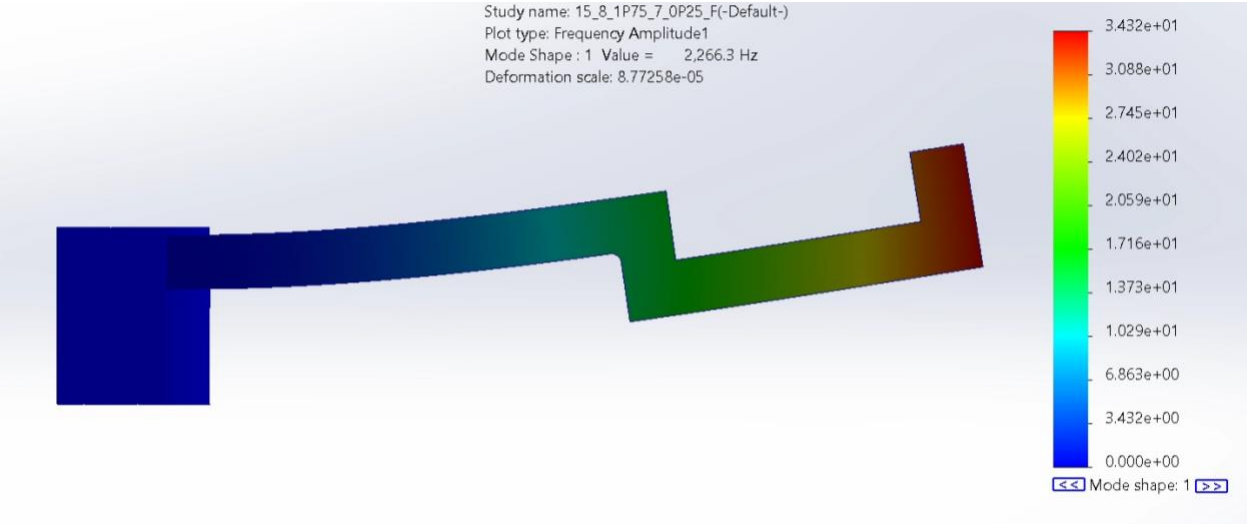


Figure 8. First mode shape (frequency amplitude) of the best-case governor leg. Fundamental natural frequency 2,266.3 Hz.

It must be noted that the requirement on the fundamental frequency was that it be below 2,100 Hz, whereas the computed value of 2,266.3 Hz is approximately 8% above this target. This was the single criterion that proved most difficult to meet simultaneously with the other constraints. As shown in the parametric design table (Table 5 below), every iteration that successfully drove the frequency below 2,100 Hz required either making the legs longer, thinner, or both — which then violated either the maximum stress criterion (case L1=13.5, L2=7.25, t=1.25, exceeding 2,200 MPa) or pushed displacement above the 1.25 mm target. The selected configuration represents the best feasible compromise: it satisfies stress, displacement, mass, and packaging constraints simultaneously while landing closest to (though slightly above) the frequency requirement. Acceptance of this design therefore depends on whether the frequency target is treated as a hard or soft constraint by the design authority.

4.4 Design Iteration Summary

Table 5 summarizes the principal design iterations explored during the parametric study, with the selected best-case design highlighted. Cells in green indicate that the corresponding criterion is satisfied; cells in red indicate violation. The selected configuration is the only one in the table that simultaneously satisfies stress, displacement, and packaging constraints with the smallest frequency overshoot.

Table 5. Parametric design iteration summary. Best case highlighted.

L1	L2	t	W	d	σ (MPa)	f (Hz)	UZ (mm)	Max (mm)	X_tot
15.25	8.25	1.75	7	0.25	1814	2213.3	1.328	31.78	25.25
15.25	8.5	1.75	7	0.25	1877	2188.5	1.38	32.08	25.5
15	8	2	7	0.75	1480	2492.7	0.8428	31.05	25
14.5	8.25	1.75	7	0.25	1770	2296.2	1.196	32.08	24.5
15	**8**	**1.75**	**7**	**0.25**	**1733**	**2266.3**	**1.234**	**31.19**	**24.75**
13.25	9	1.75	7	1	1790	2380.4	1.107	30.32	24
13.75	7.25	2	7	0.75	1345	2792.2	0.6211	28.84	23
13.5	7	1.5	7	0.75	1863	2352	1.246	28.47	22
13.5	7	1.5	7	1.5	1774	2360.3	1.242	28.47	22
13.5	7.25	1.25	7	0.75	2247	2079.2	2.03	29.26	22
12.5	6.75	1.25	7	1	1970	2252.2	1.58	27.32	20.5

4.5 Summary of Best-Case Results

The combined results for the selected best-case design are summarized in Table 6 alongside the corresponding allowable values.

Table 6. Best-case results vs. design criteria.

Result	Computed	Allowable	Status
Maximum von Mises Stress	1,793 MPa	2,200 MPa	**PASS**

Axial Deflection (UZ)	1.234 mm	1.25 mm	**PASS**
Total Radial Extent (X_tot)	24.75 mm	32 mm (housing)	**PASS**
Fundamental Frequency	2,266.3 Hz	< 2,100 Hz	**REVIEW**
Mass	3.87 g	—	—

The mass of the part was computed from the model as 3.87 g. This is consistent with the goal of using as little material as possible (criterion n) while remaining strong enough to survive the operating load. With a mass of just under 4 grams per leg-shaft assembly, the governor is light enough that gyroscopic and inertial effects on the larger system will be minimal.

5. Final Design

5.1 Final Dimensions

The final design was selected from the parametric study as the configuration that best satisfies the greatest number of design constraints simultaneously. The chosen geometry corresponds to row 5 of Table 5, with the following dimensions, all confirmed as multiples of 0.25 mm in accordance with the manufacturing requirement:

Selected from row 5 of table 5:

15	**8**	**1.75**	**7**	**0.25**	**1733**	**2266.3**	**1.234**	**31.19**	**24.75**

The total horizontal extent of the leg is $x_{tot} = L_1 + L_1 + t = 15 + 8 + 1.75 = 24$, which satisfies the upper bound of 26.75 mm derived from the housing constraint. The width W = 7.00

mm is the maximum feasible value given the local geometric restriction identified during CAD modeling, and was held at this value to maximize the leg's cross-sectional area and load-carrying capacity in accordance with criterion (n).

5.2 Leg Location on the Shaft

The governor consists of four legs arranged with cyclic symmetry about the central shaft axis, spaced 90° apart. This arrangement was enforced in the FEA model using the symmetry fixture condition, which constrains the relevant faces to produce identical deformation across all four legs under the applied centrifugal load. The symmetry fixture is visible in Figure 2 of Section 2.2, where the pattern of yellow ghost bodies confirms the four-fold arrangement. Because all legs are geometrically identical and loaded identically under pure rotation, each leg experiences the same stress, displacement, and frequency response as reported in Section 4.

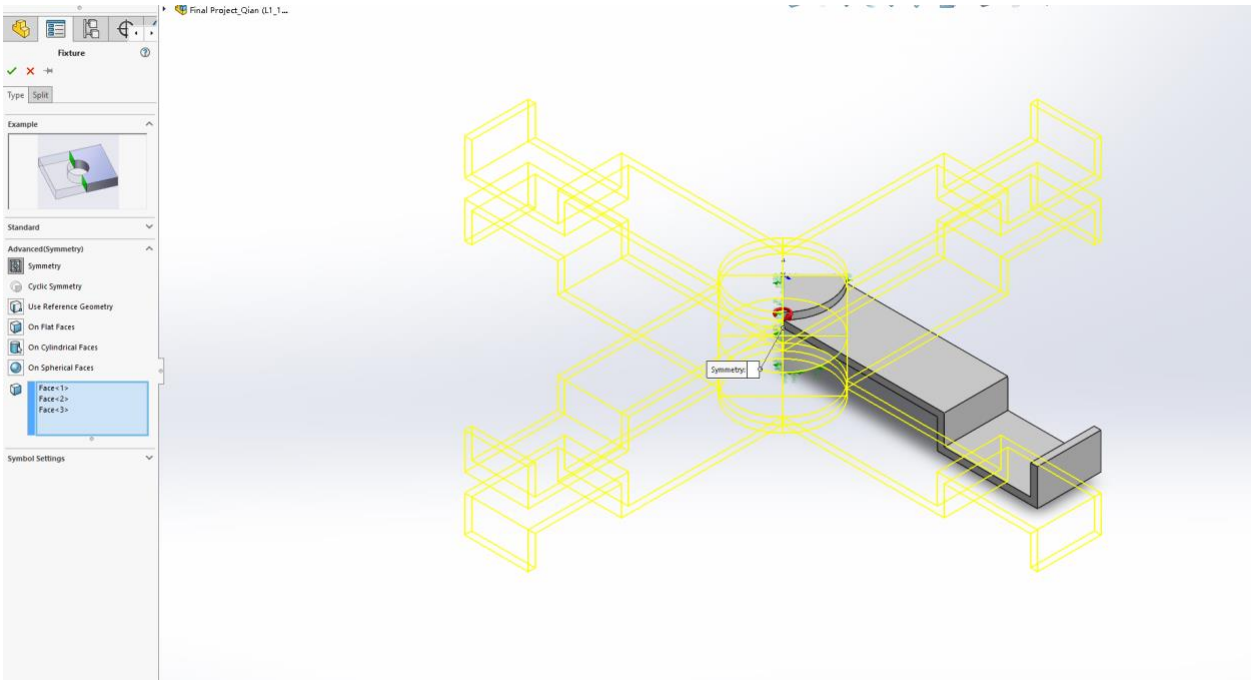


Figure 9: Legs are geometrically identical and loaded identically

5.3 Housing Fit Confirmation — Undeformed

In the undeformed configuration, the radial distance from the shaft centerline to the outermost corner of the leg is given by:

$$r_{undeformed} \approx \sqrt{(5 + L_1 + L_2 + t)^2 + \left(\frac{W}{2}\right)^2}$$

Substituting the final dimensions:

$$r_{undeformed} \approx \sqrt{(5 + 24.75)^2 + (3.5)^2} \approx 29.96$$

Since 29.96 mm < 32 mm, the governor fits comfortably within the annular housing in the undeformed state, with a radial clearance of approximately 2.04 mm.

5.4 Housing Fit Confirmation — Deformed

After applying the centrifugal load of 7,000 rad/s, the FEA static analysis yields a maximum outward tip displacement of U_z = 1.234 mm. The deformed maximum radial extent is approximated as:

$$r_{max} \approx \sqrt{(5 + L_1 + L_2 + t)^2 + \left(\frac{W}{2}\right)^2} + U_z$$

$$r_{max} \approx 29.96 + 1.234 \approx 31.19mm$$

Since 31.19 mm < 32 mm, the governor also fits within the housing in the fully deformed state, with a remaining radial clearance of approximately 0.81 mm. This confirms that criterion (o) is satisfied both statically and under operating load. It should be noted that this radial clearance is relatively tight, approximately 0.81 mm, which underscores why maximizing X_{tot} beyond 26.75 mm was not feasible without risking housing contact after deformation.

6. Conclusion

This project involved the design and analysis of a centrifugal governor leg fabricated from MIRAGE 250 CVM, constrained to fit within a fixed annular housing of radius 32 mm while satisfying requirements on von Mises stress, axial deflection, natural frequency, and overall radial extent. A parametric design study was conducted over the geometric variables L1, L2, t, W, and d, with all dimensions restricted to 0.25 mm increments, and finite element analysis was performed in SolidWorks Simulation to evaluate each candidate configuration.

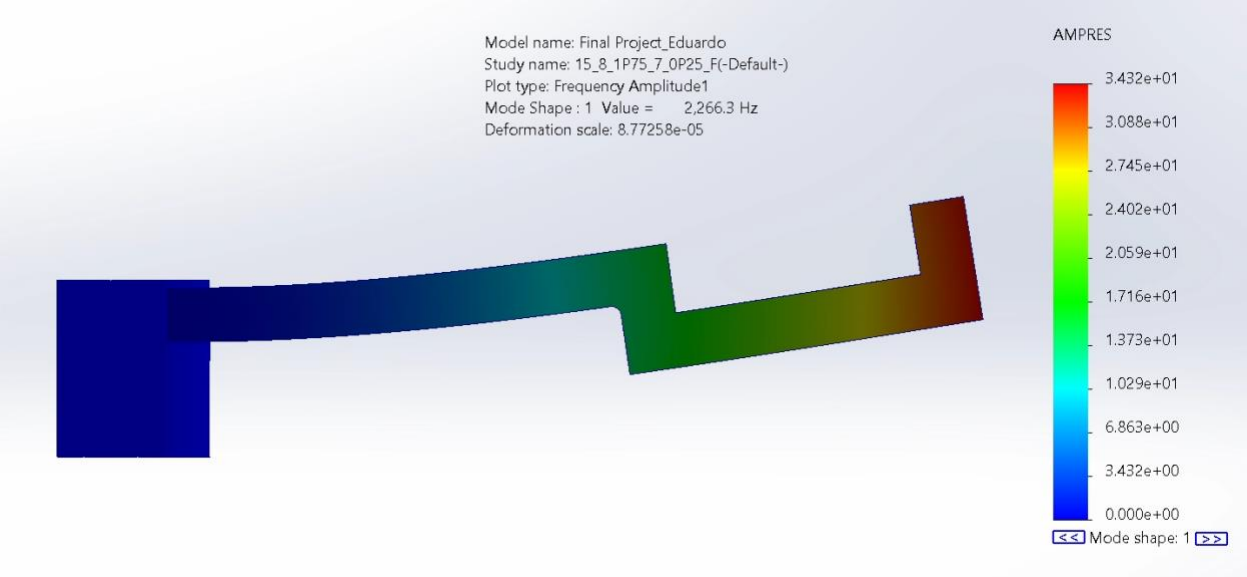
A mesh convergence study was completed prior to finalizing any results, confirming that the chosen mesh, with a maximum element size of 0.53 mm and minimum of 0.106 mm using curvature-based refinement — produces a stress prediction that is stable to within ±20 MPa across successive refinements. The final mesh contained 57,953 nodes and 38,320 elements, with 99.6% of elements having an aspect ratio below 3 and zero distorted elements, giving high confidence in the numerical results.

The selected best-case design has dimensions L1 = 15 mm, L2 = 8 mm, t = 1.75 mm, W = 7 mm, and d = 0.25 mm. This configuration satisfies three of the four primary performance constraints. The maximum von Mises stress of 1,793 MPa remains below the allowable yield strength of 2,200 MPa, providing a safety factor of 1.23. The maximum axial tip deflection of 1.234 mm falls just within the 1.25 mm single-side limit corresponding to the 2.5 mm total deflection budget. The deformed radial extent of 31.19 mm clears the 32 mm housing with 0.81 mm to spare.

The one constraint that could not be simultaneously satisfied is the natural frequency requirement. The fundamental bending frequency of the selected design is 2,266.3 Hz, approximately 8% above the 2,100 Hz target. As demonstrated through the parametric iteration table, every geometry that successfully brought the frequency below 2,100 Hz did so by increasing leg length or reducing thickness, which in every case drove either the stress above 2,200 MPa or the deflection above the 1.25 mm limit. The selected configuration represents the closest feasible approach to full constraint satisfaction given the material, geometric, and discretization requirements imposed by the project.

In summary, the design process revealed a fundamental conflict between the frequency constraint and the stress and deflection constraints for this material and housing geometry. Future work could explore alternative materials with a higher stiffness-to-density ratio, which would allow a lower natural frequency to be achieved without a corresponding increase in deflection or stress. Additionally, a non-uniform cross-section or tapered leg geometry could be investigated as a means of redistributing stiffness to reduce frequency while keeping peak stress within bound

Appendix



Curve Based Mesh					
	Max Elemnt Size	Min elemnt Size [mm]	min # of elemnts in circle	elemnt size groeth ratio	Stress [Mpa]
	1.5	0.3	8	1.4	1867.11
	1.25	0.25	8	1.4	1724.95
	1.1	0.22	8	1.4	1802.85
	1	0.2	8	1.4	1799.67
	0.9	0.18	8	1.4	1786.46
	0.8	0.16	8	1.4	1720
	0.75	0.15	8	1.4	1763
	0.7	0.14	8	1.4	1756
	0.65	0.13	8	1.4	1763
	0.6	0.12	8	1.4	1723
	0.55	0.11	8	1.4	1779
	0.54	0.108	8	1.4	1773
	0.53	0.106	8	1.4	1793
	0.52	0.104	8	1.4	1796
	0.5	0.1	8	1.4	1794
	0.45	0.09	8	1.4	1797

L1	L2	t	W	d	vonM stress [Mpa] < 2200	Frequency [Hz] < 2100	UZ Dis < 1.25 [mm]	Max Dis < 32 [mm]	X_tot [mm]			UZ Dis < 0.125 [mm]
16	9	1	7	0.75	3527	1613.6	6.51	37.70695498	26		16_9_1_7_0P75	0.651
16	9	1.25	7	0.75	2933	1756	3.874	35.31938917	26.25			0.3874
15.75	9	2	7	0.75	1432	2281.3	1.09	33.03233085	26.75			0.109
15.75	8.5	2	7	0.75	1362	2333.7	1.019	32.46438917	26.25		15P75_8P5_2_7_0P75	0.1019
15.75	9	1.5	7	0.75	2183	1930.9	2.395	33.84038917	26.25			0.2395
15.25	9.5	1.5	7	0.75	2241	1932.6	2.449	33.89438917	26.25		15P25_9P5_1P5_7_0P75	0.2449
16	8.75	1.5	7	0.75	2227	1929.9	2.424	33.86938917	26.25			0.2424
15.75	9	1.75	7	0.75	1953	2101.6	1.624	33.31784798	26.5		15P75_9_1P75_7_0P75	0.1624
15.5	9.25	1.75	7	0.75	1820	2101.8	1.605	33.29684798	26.5			0.1605
16	8.75	1.75	7	1	1775	2101.3	1.592	33.28584798	26.5		16_8P75_1P75_7_1	0.1592
16	8.75	1.75	7	0.5	3088	2096.6	1.596	33.28984798	26.5			0.1596
16	9	1.75	7	0.75	3084	2079.4	1.653	33.59533085	26.75			0.1653
16	8.75	2	7	0.75	1434	2280.3	1.093	33.03533085	26.75			0.1093
15.75	9	1.75	7	1	1833	2104.3	1.623	33.31684798	26.5		15P75_9_1P75_7_1	0.1623
15.5	9.25	1.75	7	1	1873	2105.3	1.604	33.29784798	26.5		15P5_9P25_1P75_7_1	0.1604
15.5	9	1.75	7	1	1817	2127.1	1.548	32.99138917	26.25		15P5_9_1P75_7_1	0.1546
15.75	8.75	2	7	1	1377	2310.9	1.057	32.75084798	26.5		15P75_8P75_2_7_1	0.1057
15.5	9	2	7	1	1460	2312.2	1.075	32.76884798	26.5		15P5_9_2_7_1	0.1075
16	8.5	1.75	7	1	1666	2125.1	1.534	32.97938917	26.25		16_8P5_1P75_7_1	0.1534
16	8.5	1.75	7	0.75	1703	2122.6	1.535	32.98038917	26.25		16_8P5_1P75_7_0P75	0.1535
15.75	8.75	1.75	7	0.75	1766	2123.7	1.542	32.98738917	26.25		15P75_8P75_1P75_7_0P75	0.1542
15.75	8.5	1.75	7	0.75	1688	2146.3	1.485	32.68195498	26		15P75_8P5_1P75_7_0P75	0.1485
15.25	8.75	2	7	0.75	1341	2364.1	0.9886	32.18555498	26		15P25_8P75_2_7_0P75	0.09886
16	8.5	2	6	0.75	1399	2310.2	1.052	32.89453466	26.5		16_8P5_2_6_0P75	0.1052
16	8.75	1.75	6	0.75	1772	2104.1	1.592	33.23453466	26.5		16_8P75_1P75_6_0P75	0.1592
15.75	8.5	2	6	0.75	1419	2338.4	1.018	32.41166974	26.25		15P75_8P5_2_6_0P75	0.1018
15.75	8.5	2	6	0.5	1447	2332.8	1.019	32.41266974	26.25		15P75_8P5_2_6_0P5	0.1019
15.5	8.5	2	6	0.75	1402	2367.5	0.9837	32.128523	26		15P5_8P5_2_6_0P75	0.09837
16	8.75	2	6	0.75	1353	2284.4	1.091	32.98241734	26.75		16_8P75_2_6_0P75	0.1091
15.5	9.25	2	6	0.75	1506	2286.6	1.099	32.99041734	26.75		15P5_9P25_2_6_0P75	0.1099
15.5	9	2	6	0.75	1474	2312.8	1.06	32.70253466	26.5		15P5_9_2_6_0P75	0.106
15.5	9	2	7	0.75	1629	2315.4	1.05	32.74384798	26.5	Fillet	15P5_9_2_7_0P75	0.105
15.25	9.25	1.75	7	1	1439	2128.5	1.541	32.98638917	26.25		15P25_9P25_1P75_7_1	0.1541
15.5	8.25	1.25	5	0.75	2563	1836	3.231	33.33496645	25		15P5_8P25_1P25_5_0P75	0.3231
15	8	2	5	0.75	1483	2503.7	0.8394	30.94338645	25		15_8_2_5_0P75	0.08394
15	8	2	4	0.75	1447	2511.7	0.8358	30.90239276	25		15_8_2_4_0P75	0.08358
15	8	2	7	0.75	1480	2492.7	0.8428	31.04627662	25			0.08428

15	8.25	1.75	7	0.25	1794	2240	1.283	31.48647662	25		15_8P25_1P75_7_0P25	
15.25	8.25	1.5	7	0.75	2215	2042.5	1.975	32.17647662	25		15P25_8P25_1P5_7_0P75	0.1975
14.5	8	1.5	7	0.75	2184	2034.2	1.975	31.18544334	24		14P5_8_1P5_7_0P75	0.1975
14.5	7.75	1.75	7	0.75	1679	2369.9	1.098	30.30844334	24		14P5_7P75_1P75_7_0P75	0.1098
14.5	7.75	1.75	7	0.25	1681	2353.1	1.104	30.31444334	24		14P5_7P75_1P75_7_0P25	
14	7.5	1.5	7	0.75	1983	2232.1	1.464	29.68190212	23		14_7P5_1P5_7_0P75	0.1464
14	7.75	1.25	7	0.75	2384	1986.5	2.384	30.60190212	23		14_7P75_1P25_7_0P75	0.2384
13.75	7.25	2	7	0.75	1345	2792.2	0.6211	28.83900212	23		13P75_7P25_2_7_0P75	0.06211
12.5	8.75	1.75	7	0.25		2492.2			23		12P5_8P75_1P75_7_0P25	
13.5	7	1.5	7	0.75	1863	2352	1.246	28.47190678	22		13P5_7_1P5_7_0P75	0.1246
13.5	7.25	1.25	7	0.75	2247	2079.2	2.03	29.25590678	22		13P5_7P25_1P25_7_0P75	0.203
13.5	7.25	1.5	7	0.25		2305.6			22.25		13P5_7P25_1P5_7_0P25	
13.25	7	1.75	7	0.75	1430	2653	0.8049	28.03080678	22		13P25_7_1P75_7_0P75	
13.25	9	1.75	7	1	1790	2380.4	1.107	30.31744334	24		13P25_9_1P75_7_1	0.1107
12	6.5	1.5	7	1	1556	2661	0.8909	26.13471112	20		12_6P5_1P5_7_1	0.08909
12.25	6.5	1.25	7	1	1920	2315.7	1.448	26.69181112	20		12P25_6P5_1P25_7_1	0.1448
11.25	7.75	1	7	1	2671	2004.4	2.573	27.81681112	20		11P25_7P75_1_7_1	
12.5	6.75	1.5	7	1	1677	2536.8	1.014	27.00077548	20.75		12P5_6P75_1P5_7_1	0.1014
12.5	6.75	1.25	7	1	1970	2252.2	1.58	27.31907535	20.5		12P5_6P75_1P25_7_1	0.158
12.5	6.75	1	7	1	2304	2080.7	2.141	27.63242013	20.25		12P5_6P75_1_7_1	
11.5	6.5	1	7	1	2362	2107.2	2.212	26.46586567	19		11P5_6P5_1_7_1	
10.5	6.5	1	7	1	2180	2235.5	1.779	25.04378025	18		10P5_6P5_1_7_1	
9.5	6.5	1	7	1	2047	2387.6	1.46	23.73666941	17		9P5_6P5_1_7_1	
9.5	5.5	1	7	1	1817	2562	1.186	22.47566886	16		9P5_5P5_1_7_1	